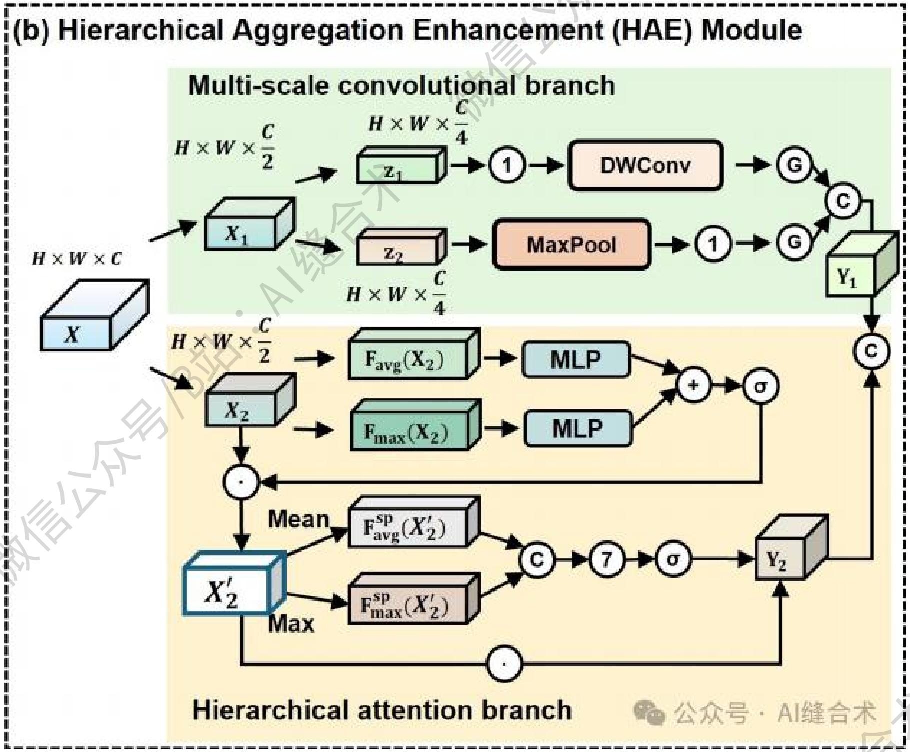
核心速览



论文题目： EccoMamba: Enhanced Cross-hierarchical Continuity Orthogonal Mamba for Medical Image Segmentation

中文题目： EccoMamba：用于医学图像分割的增强型跨层级连续性正交 Mamba 算法

论文链接：

https://x.sci-hub.org.cn/target?link=https://ojs.aaai.org/index.php/AAAI/article/view/38109

所属单位： 武汉理工大学等

核心速览：

提出 EccoMamba，一种 U 形编码器-解码器框架，通过（层次聚合增强，HAE）模块和（结构连续性正交模块，SCO）模块，解决现有 Mamba 架构在医学图像分割中存在的分层解剖特征捕捉不足和空间依赖性建模问题，提升分割准确性和结构保真度。

https://mp.weixin.qq.com/s/jG9lhTLHezEkrF22Xd3YDQ

```
HAE(
  (Maxpool): MaxPool2d(kernel_size=3, stride=1, padding=1, dilation=1, ceil_mode=False)
  (proj_maxpool): Conv2d(32, 64, kernel_size=(1, 1), stride=(1, 1))
  (mid_gelu_maxpool): GELU(approximate='none')
  (attention_conv): Conv2d(16, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
  (attention_proj): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=32, bias=False)
  (mid_gelu_attention): GELU(approximate='none')
  (channel_attention): ChannelAttentionModule(
    (avg_pool): AdaptiveAvgPool2d(output_size=1)
    (max_pool): AdaptiveMaxPool2d(output_size=1)
    (fc1): Conv2d(32, 2, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (relu1): ReLU()
    (fc2): Conv2d(2, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (sigmoid): Sigmoid()
  )
  (spatial_attention): SpatialAttentionModule(
    (conv1): Conv2d(2, 1, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), bias=False)
    (sigmoid): Sigmoid()
  )
  (dwconv_conv): Conv2d(16, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
  (dwconv_proj): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=32, bias=False)
  (mid_gelu_dwconv): GELU(approximate='none')
  (conv_fuse): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=128, bias=False)
  (proj): Conv2d(128, 64, kernel_size=(1, 1), stride=(1, 1))
  (proj_drop): Dropout(p=0.0, inplace=False)
)
input_tensor_shape  : torch.Size([2, 64, 32, 32])
output_tensor_shape : torch.Size([2, 64, 32, 32])

哔哩哔哩/微信公众号：AI缝合术，独家整理！
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